

STAT/MA 41600

In-Class Problem Set #9: September 14, 2026

1. Alice rolls a 6-sided dice (numbered 1 to 6), and Bob rolls a 20-sided dice (numbered 1 to 20). What is the probability the results of the dice rolls agree?

2. Alice flips a fair coin until the first time that she gets heads, and then she stops. Bob also flips a fair coin until the first time that he gets heads, and then he stops. What is the probability that they both flipped the coin the same number of times?

3. Suppose that $P(Y = y) = pq^{y-1}$ for integers $y \geq 1$, where $q = 1 - p$.

3a. Find the probability that Y is even, i.e., that Y is a multiple of 2.

3b. Find the probability that Y is a multiple of 3.

3c. Find the probability that Y is a multiple of 4.

3d. Fix an integer $r \geq 1$. Find the probability that Y is a multiple of r .

4. Consider two independent discrete random variables, X and Y , with probability mass functions $f_X(x) = (1/2)^{x+1}$ for $x \geq 0$ and $f_Y(y) = (e^{-2})(2^y)/(y!)$ for $y \geq 0$.

What is $P(X = Y)$?

5. Consider two independent discrete random variables, X and Y , with probability mass functions $f_X(x) = (3/5)^{x-1}(2/5)$ for $x \geq 1$ and $f_Y(y) = (2/7)^{y-1}(5/7)$ for $y \geq 1$.

What is $P(X \leq Y)$?